

POCKET PRODUCER: SESSION-CONDITIONED RETRIEVAL AS USER-CONTROLLED PROPOSAL

Chuling Li

University of Nottingham

scycl14@nottingham.ac.uk

ABSTRACT

Music continuation need not coincide with audio similarity: the loop that best matches a session is not necessarily the one that best moves it forward. Pocket Producer is a working prototype integrated into the Audiotool DAW and designed around this distinction. It fingerprints the live session, ranks candidate loops using pack co-membership as cold-start weak supervision, and presents the results through an inspectable and reversible proposal workflow. Offline studies on a fixed held-out split show that the apparent advantage of the learned ranker is sensitive to label construction, negative regime, and embedding representation. These observations motivate a design stance in which retrieval scores generate *proposals* that musicians can inspect, audition, insert, and undo, rather than final judgments about what should come next.

1. INTRODUCTION

When a musician returns to an unfinished loop-based project, the question is not “what sounds similar?” but “what would be useful as the next step?” Loop compatibility [1–4], stem coherence [5–7], session-conditioned generation [8], and reversible co-creation [9] all address parts of this problem. We define **continuation utility** as the property that a candidate helps the musician make progress in the current session; unlike pairwise audio similarity, it is directional and conditioned on the evolving session context.

We contribute: (1) a problem formulation that separates continuation utility from similarity and compatibility; (2) a working prototype in a production DAW, engineered for revisability: live session fingerprinting over the Nexus SDK, per-signal evidence, single-transaction insertion whose undo also retracts the candidate from the ranking context, and feedback logging; (3) a tiered negative-regime evaluation protocol with a fixed held-out split that characterizes the sensitivity of weakly supervised ranking to label construction, negative regime, and embedding representation.

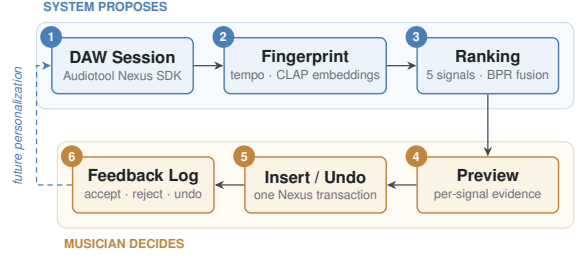


Figure 1. Pocket Producer pipeline: the system proposes (steps 1–3); the musician decides (steps 4–6).

2. SYSTEM

2.1 Session Fingerprinting

The live Audiotool session (Figure 1) is read via the Nexus SDK, which exposes the project’s tracks, regions, and referenced audio samples. The system extracts the project tempo from the document metadata and collects up to eight unique audio samples referenced by regions in the arrangement. Each sample is embedded with CLAP [10]; the resulting vectors form the session’s timbral fingerprint. When no session audio is available (empty project or no sample-backed regions), the system falls back to non-audio signals only.

2.2 Multi-Signal Ranking

Five signals score each FSLD [11] candidate against the session fingerprint: (i) mean-pooled and (ii) max-pooled CLAP cosine similarity to session samples, (iii) tempo proximity, (iv) key compatibility, and (v) tag Jaccard overlap between candidate tags and tags from the session’s audio fragments. Writing $\phi(s, x) \in \mathbb{R}^5$ for these signals given session s and candidate x , the fusion scores each candidate as $\sigma(w^\top \phi(s, x))$, with w trained by BPR [12] on pack co-membership weak labels (Section 3):

$$\mathcal{L}_r(w) = -\mathbb{E}_{\mathcal{D}_r} [\log \sigma(w^\top (\phi(s, x^+) - \phi(s, x^-)))], \quad (1)$$

where the expectation is over triples (s, x^+, x^-) and r indexes the negative regime from which x^- is drawn. A hand-tuned *rules* baseline over non-audio session signals serves as the default ranker; the fusion is selectable and falls back to rules when no session audio is available.

2.3 Proposal Workflow

The proposal workflow keeps creative authority with the musician: every recommendation can be inspected, audi-



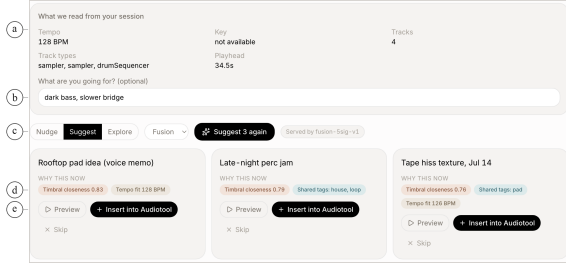


Figure 2. Continuation panel (screenshot). (a) Session fingerprint via the Nexus SDK; (b) text intent; (c) intensity and ranker; (d) evidence chips; (e) audition, insert, skip. Candidates come from the musician’s captured fragments.

tioned, inserted, and undone. *Inspect*: each candidate carries per-signal evidence, reason chips naming the signals that favored it (e.g., timbral closeness to session regions, tempo fit, shared tags), so the musician can judge whether those reasons apply to the current goal; signals that contribute nothing emit no evidence. *Audition*: candidates can be previewed before insertion without modifying the document. *Insert*: an accepted candidate enters the arrangement via a single Nexus transaction. *Undo*: the most recent insertion can be reverted, which also retracts the candidate’s tags from the ranking context, so subsequent recommendations are not biased by the undone choice. Preview, insertion, rejection, and undo events are logged for future personalization (not yet evaluated).

2.4 Deployed Model

The deployed ranker follows the same five-signal recipe but is a separate production model with four stated deltas: MS-CLAP embedding space (offline evaluation uses LAION-CLAP Music); source-weighted BPR ($\text{hard_similar} \times 2.0$, mean over 5 seeds); key constant zero online (Audiotool documents lack key-signature metadata); tag context drawn from session fragments rather than pack siblings.

3. COLD-START WEAK SUPERVISION

At launch no interaction data exists, so the fusion is trained on proxy supervision that predates any user: pack co-membership. From the FSLD [11] we kept only sounds with a true Freesound pack (1,797 items, 605 packs), discarding items grouped only by uploader. Each eligible pack with at least three items yields one leave-one-out query per item, with same-pack siblings as positives and cross-pack items as negatives in three difficulty tiers (*easy*: tempo/key mismatch; *hard_tempo_key*: compatible tempo and key; *hard_similar*: highest cosine to the positive from another pack). A 121-source held-out split (seed 20260810) was fixed in version control before any evaluation ran. Development uses only the 484 train sources; the 171 with ≥ 3 items form queries (981 examples), the rest contribute negatives only. LAION-CLAP Music [13] (512-dim) provides the embedding backbone.

On the train side (mean over 5 source-grouped splits, each training on $\sim 75\%$ of train sources for 300 epochs, no

Negative regime	LAION-CLAP	MS-CLAP, sw-BPR
easy	+0.2 [−0.5, +0.9]	+0.1 [−1.2, +1.4]
hard_tempo_key	+0.0 [−0.8, +0.5]	−0.1 [−2.2, +2.0]
hard_similar	+2.1 [−1.0, +5.3]	+9.9 [+4.4, +15.4]
overall	+0.6 [−0.4, +1.6]	+2.6 [+1.0, +4.2]

Table 1. Held-out fusion—cosine delta (pp) with bootstrap 95% CIs, per negative regime and embedding/training configuration. The apparent gain concentrates in *hard_similar*; the 7.8 pp cross-configuration swing shows sensitivity to embedding and training choices.

early stopping), frozen cosine reaches 0.929 overall and 0.814 on *hard_similar*; five-signal fusion reaches 0.925 and 0.813. Under pack-only labels the fusion advantage is gone: -0.03 pp on *hard_similar*, and fusion trails cosine on *hard_tempo_key* by 2.1 pp on average (negative in all 5 splits).

4. OBSERVATIONS

Three lines of evidence characterize the sensitivity of the cold-start proxy. **Label sensitivity.** An earlier corpus treated same-uploader items as co-members (26% of examples, 949 sources); fusion appeared $+4.0$ pp over frozen cosine on *hard_similar*. Restricting to true pack co-membership erases the gain (Section 3). **Regime dependence.** Leave-one-signal-out ablation on the dev set shows tempo helps on *hard_similar* (-0.96 pp when removed) but hurts on *hard_tempo_key* ($+0.68$ pp); signals switch sign across negative regimes. In the terms of Eq. (1), the learned optimum w^* depends on the training mixture over regimes $\mathcal{D}_r$ rather than defining a regime-invariant ranking. **Held-out uncertainty.** An initial held-out run trained each seed on a 75% internal subset of train sources; after identifying this as a protocol mismatch, we retrained on all 484 train sources and re-evaluated. Table 1 reports the corrective results; the held-out split is therefore exploratory rather than confirmatory. Under LAION-CLAP the *hard_similar* advantage is $+2.1$ pp with a bootstrap 95% CI crossing zero; under MS-CLAP with source-weighted BPR (a sensitivity re-analysis on the same split) it is $+9.9$ pp, a 7.8 pp cross-configuration swing. *Hard_tempo_key* shows near-zero deltas in both configurations, with CIs crossing zero (the initial run showed a negative *hard_tempo_key* delta under LAION-CLAP that disappeared after the correction).

5. CONCLUSION

Pocket Producer is a working prototype that operationalizes session-conditioned retrieval as a user-controlled proposal workflow in a production DAW: live fingerprinting, per-signal evidence, reversible insertion, and feedback logging. The default ranker is a transparent rules baseline requiring no session audio; the fusion is selectable. Future work includes evaluating whether the proposal workflow improves creative outcomes, personalization from logged feedback, and ranking objectives that are more robust across negative regimes.

6. ACKNOWLEDGEMENTS

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7. AI USAGE STATEMENT

Claude Code (Anthropic) was used for engineering implementation, test scaffolding, statistical scripting, prose editing, result auditing (numerical cross-checks between artifacts and manuscript), literature searching, and narrative restructuring. All experimental design, analysis decisions, claim selection, and final manuscript content are the sole responsibility of the author.

8. REFERENCES

- [1] B.-Y. Chen, J. B. L. Smith, and Y.-H. Yang, “Neural loop combiner: Neural network models for assessing the compatibility of loops,” in *Proceedings of the 21st International Society for Music Information Retrieval Conference (ISMIR)*, 2020.
- [2] S. Lattner, “SampleMatch: Drum sample retrieval by musical context,” in *Proceedings of the 23rd International Society for Music Information Retrieval Conference (ISMIR)*, 2022.
- [3] X. Yi and M. Barthet, “A generative framework for composition-aware loop recommendation in music production: Drum2Bass use case,” in *Proceedings of the Sound and Music Computing Conference (SMC)*, 2024.
- [4] S. Roy Chaudhuri, S. P. Madupu, K. T. Guru Sai, and V. Jamwal, “LoopMatcher: Proof-of-concept for AI-assisted music loop search,” in *Proceedings of the 15th International Conference on Computational Intelligence in Music, Sound, Art, and Design (EvoMUSART)*, 2026.
- [5] R. Ciranni, G. Mariani, M. Mancusi, E. Postolache, G. Fabbro, E. Rodolà, and L. Cosmo, “COCOLA: Coherence-oriented contrastive learning of musical audio representations,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [6] J. Nistal, M. Pasini, C. Aouameur, M. Grachten, and S. Lattner, “Diff-A-Riff: Musical accompaniment co-creation via latent diffusion models,” in *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, 2024.
- [7] A. Riou, A. Gagnere, G. Hadjeres, S. Lattner, and G. Peeters, “Zero-shot musical stem retrieval with joint-embedding predictive architectures,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [8] Q. He, G. Xia, and Z. Wang, “TOMI: Transforming and organizing music ideas for multi-track compositions with full-song structure,” in *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR)*, 2025.
- [9] A. C. Elkins, S. Singh, A. Kieback, S. Blankenship, U. P. Amadasun, and A. Chadha, “DAWZY: A new addition to AI powered “human in the loop” music co-creation,” in *NeurIPS Workshop on AI for Music*, 2025.
- [10] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, “CLAP: Learning audio concepts from natural language supervision,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [11] A. Ramires, F. Font, D. Bogdanov, J. B. L. Smith, Y.-H. Yang, J. Ching, B.-Y. Chen, Y.-K. Wu, W.-H. Hsu, and X. Serra, “The Freesound Loop Dataset and annotation tool,” in *Proceedings of the 21st International Society for Music Information Retrieval Conference (ISMIR)*, 2020.
- [12] S. Rendle, C. Freudenthaler, Z. Gantner, and L. Schmidt-Thieme, “BPR: Bayesian personalized ranking from implicit feedback,” in *Proceedings of the 25th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2009.
- [13] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, “Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation,” in *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.